

M.Tech in CSE (Data Centre Systems Engineering) · AY 2026–2028 · Semester I

ACADEMIC COUNCIL APPROVED

ADVANCE DATA CENTRE AND CLOUD INFRASTRUCTURE ENGINEERING (ADC & CIE)

09 JUN 2026 · REGULAR

Course Code: **CDS40010** | Category: Program Major (PM) | Credits: **4 (L:3, T:0, P:2)**

Teaching Scheme	Lectures	Tutorial	Practical	Total Load	Credits	Assessment Scheme (Pattern TL3)
Weekly Hours	3 Hrs/wk	—	2 Hrs/wk	5 Hrs/wk	4 Credits	CCA:20 (10+10), MT:25, LCA:15 (5+5+5), TE:40

Prerequisites: Computer Networks, Operating Systems, Cloud Computing Fundamentals, Intro to Data Centre Technologies, Electrical & Mechanical Fundamentals.

COURSE OBJECTIVES (COB)

1. Understand architecture, design standards (TIA-942), & Tier classifications.
2. Examine cloud infrastructure models (IaaS, PaaS, SaaS, CaaS) & multi-cloud orchestration.
3. Analyse electrical distribution, UPS topologies, redundancy, & PUE energy efficiency.
4. Evaluate thermal cooling (CRAC/CRAH, liquid cooling, economizers, airflow management).
5. Design high-speed network topologies, Software-Defined Networks (SDN) & spine-leaf fabrics.
6. Apply AI/ML-driven DCIM tools for telemetry, predictive maintenance & capacity planning.

COURSE OUTCOMES (CO)

- CO1: Recall & explain TIA-942/Uptime Institute tier architecture & design standards.
CO2: Architect secure, scalable, & cost-optimised cloud infrastructures (IaaS/PaaS).
CO3: Calculate power distribution schemes, UPS capacity, & PUE/WUE efficiency metrics.
CO4: Design high-availability spine-leaf network topology & SDN data centre interconnects.
CO5: Evaluate AI/ML-based DCIM platforms for telemetry, fault prediction, & reporting.
CO6: Formulate complete end-to-end green data centre infrastructure engineering design.

THEORY SYLLABUS (6 UNITS · 42 INSTRUCTIONAL HOURS)

UNIT 1: Fundamentals of Data Centre Infrastructure

[6h]

- Introduction to Data Centres: Evolution, enterprise, hyperscale, colocation, & edge.
- Facility Infrastructure: White space, grey space, structural floor loading, raised floor.
- Standards & Tiers: TIA-942 and Uptime Institute Tier I, II, III, IV classifications.
- Fire Protection & Physical Security: Zoning, perimeter defence, access control.
- Data Centre Design: Site selection, seismic zoning, resilience & modular scalability.

UNIT 2: Electrical Infrastructure Engineering

[6h]

- Utility Power Path: HV/LV substations, transformers, MLVS, bus ducts, ATS & STS switches.
- Backup Power Systems: Diesel Generator (DG) sizing, paralleling, fuel autonomy, AMF panels.
- UPS Systems: Online double conversion, delta conversion, VRLA vs Li-ion battery strings.
- Redundancy Configurations: N, N+1, 2N, 2(N+1) electrical fault tolerance.
- Power Distribution: Floor PDUs, intelligent rack PDUs, earthing, bonding, lightning protection.

UNIT 3: Mechanical Infrastructure, HVAC & Thermal Mgmt

[8h]

- HVAC Fundamentals: Psychrometric chart, ASHRAE TC 9.9 thermal envelopes (A1–A4).
- Precision Cooling: CRAC (DX downflow/upflow) vs CRAH (chilled water plant loops).
- Airflow Optimization: Hot Aisle / Cold Aisle containment (HACC/CACC), perforated tiles.
- Liquid Cooling: In-row units, rear-door heat exchangers (RDHx), direct-to-chip, immersion.
- Energy Efficient Cooling: Economizers (air/water free cooling), PUE calculation.

UNIT 4: Networking, Security & Operations

[8h]

- Networking Fabric: Spine-Leaf topology, Top-of-Rack (ToR), End-of-Row (EoR), SDN.
- Structured Cabling: TIA-606-C labelling, OM4/OS2 fiber optics, Cat6A copper, patch racks.
- Physical Security: Multi-layer defence, biometric readers, man-traps, CCTV coverage.
- Fire & Life Safety: VESDA aspirating smoke detection, clean agent suppression (FM-200/Novec).
- Operations & DR: Incident response, standard SOPs, disaster recovery, business continuity.

UNIT 5: Intelligent DCIM, Automation & Sustainability

[8h]

- DCIM Platforms: Real-time telemetry, asset tracking, power chain mapping, thermal mapping.
- Building Management Systems: BACnet, Modbus, SNMP environmental sensor integration.
- AI/ML Operations: Predictive maintenance, anomaly detection, digital twin facility models.
- Green Data Centres: Renewable energy (solar PPA), PUE/WUE/CUE metrics, carbon accounting.
- Compliance & Governance: ISO 27001, SOC 2, LEED certification, statutory regulatory audit.

UNIT 6: Advanced Cloud Infrastructure & Hybrid Cloud

[8h]

- Cloud Service Models: IaaS, PaaS, SaaS, Containers as a Service (CaaS), Serverless (FaaS).
- Deployment Architectures: Public, Private, Hybrid Cloud, Multi-Cloud workload orchestration.
- Software-Defined DC: Compute virtualization (KVM/ESXi), Storage spaces, SDN overlays (VXLAN).
- Connectivity & Security: IPsec VPN tunnels, AWS Direct Connect, Azure ExpressRoute, Zero Trust.
- Infrastructure as Code: Terraform declarative provisioning, cloud monitoring & telemetry.

LIST OF LABORATORY EXPERIMENTS & PRACTICALS (10 EXPERIMENTS · 2 HRS/WEEK)

Exp	Practical Title	Practical Scope & Deliverables
1	Physical Layout & Functional Zoning	Study physical layout, security perimeters, and functional zones (white space, UPS room, cooling yard, NOC, loading bay) of a Tier III/Tier IV data centre.
2	UPS, Battery & DG Sizing Calculations	Perform IT load estimation, calculate UPS kVA capacity (with 20% headroom), battery bank AH capacity (VRLA/Li-ion), and Diesel Generator kVA sizing.
3	Electrical Single Line Diagram (SLD)	Prepare and trace an electrical Single Line Diagram (SLD) from utility substation, transformers, ATS/STS switches, UPS buses to floor PDUs and server racks.
4	Airflow Management & Aisle Containment	Design hot aisle–cold aisle containment layout (HACC/CACC), analyze perforated tile placement, and compute airflow (CFM) requirements for 5–10 kW/rack.
5	CRAC, CRAH & Precision Cooling Systems	Study CRAC (direct expansion DX) and CRAH (chilled water) precision cooling units, primary/secondary cooling loops, and calculate sensible heat removal.
6	Structured Cabling & Rack Layout	Design structured cabling architecture (TIA-606-C), overhead cable tray routing, OM4 multimode fibre and Cat6A copper patch panel termination schemes.
7	Virtual Machine Deployment on Cloud	Provision and configure Virtual Machines on cloud platforms (AWS / Azure), establish virtual network (VNet/ VPC), Network Security Groups (NSGs), and SSH/RDP access.
8	Infrastructure as Code (IaC) via Terraform	Author Terraform configuration files (.tf) to automate multi-tier cloud infrastructure provisioning (Compute, Storage, Security Groups, and Subnets).
9	DCIM Software Setup & Telemetry Monitoring	Install and configure a DCIM tool (e.g. OpenDCIM), import rack asset inventory, map power/cooling topology, and monitor real-time PUE and temperature sensors.
10	Hybrid Cloud & Disaster Recovery Architecture	Design a hybrid cloud architecture connecting on-premise data centre resources to cloud VPC via site-to-site IPsec VPN, configure automated backup and disaster recovery.

RECOMMENDED TEXTBOOKS & REFERENCE BOOKS

1. Hwaiyu Geng, 'Data Center Handbook', 2nd Edition, Wiley, 2021.
2. Mauricio Arregoces & Maurizio Portolani, 'Data Center Fundamentals', Cisco Press, 2003.
3. Thomas Erl et al., 'Cloud Computing: Concepts, Technology & Architecture', Pearson, 2013.
4. Michael J. Kavis, 'Architecting the Cloud: Service Models & Decisions', Wiley, 2018.
5. Yevgeniy Brikman, 'Terraform: Up & Running', 3rd Edition, O'Reilly Media, 2022.

ONLINE & INDUSTRY BENCHMARK RESOURCES

- The Green Grid: thegreengrid.org (PUE & DC Energy Metrics)
- Cisco Data Center: cisco.com (ACI, Spine-Leaf Architecture)
- Microsoft Learn: learn.microsoft.com (Azure Cloud Infrastructure)
- VMware SDDC: docs.vmware.com (Virtualization & SDDC)
- Cloud Native Computing Foundation: cnf.io (Kubernetes & CaaS)

EVALUATION & ASSESSMENT SCHEME (PATTERN TL3)

- Class Continuous Assessment 1 (CCA1): 10 Marks
- Lab Continuous Assessment 1 (LCA1): 5 Marks
- **Term End Examination (TE): 40 Marks (3 Hours)**
- Mid Term Exam (MT): 25 Marks
- Lab Continuous Assessment 2 (LCA2): 5 Marks
- **Total Course Weightage: 100 Marks (Passing Minimum: 50% Aggregate)**
- Class Continuous Assessment 2 (CCA2): 10 Marks
- Lab Continuous Assessment 3 (LCA3): 5 Marks

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